Summary and Recommendation

This analysis examines customer churn behavior using exploratory data analysis and visualization techniques. The dataset indicates that approximately **26.5% of customers have churned**, while **73.5% have been retained**, highlighting a moderate but business-critical churn rate.

A key finding is the strong relationship between **tenure and churn**. Customers with **tenure less than 2 months exhibit churn rates exceeding 50–60%**, making them the most vulnerable segment. In contrast, customers with tenure greater than 24 months show churn rates dropping below **15%**, indicating that **customer loyalty strengthens significantly over time**.

Demographic analysis reveals that **senior citizens have a higher churn rate (~41–42%)** compared to non-senior customers (**~23–24%**). This suggests that nearly **1 in 2 senior customers are at risk**, emphasizing the need for targeted retention strategies such as simplified services or personalized support.

Contract type emerges as one of the most influential factors:

- **Month-to-month contracts:** ~42–45% churn rate
- **One-year contracts:** ~10–12% churn rate
- **Two-year contracts:** ~2–5% churn rate

This clearly shows that customers on flexible contracts are **4–5 times more likely to churn**, reinforcing the importance of promoting long-term commitments.

Service-based analysis highlights that customers without add-on services have significantly higher churn:

- Without **Online Security / Tech Support / Backup:** ~30–40% churn
- With these services: ~15–20% churn

This indicates that **value-added services can reduce churn probability by nearly half**, making them critical for retention strategies.

Internet service type also shows a clear pattern:

- **Fiber optic users:** ~40–42% churn
- **DSL users:** ~18–20% churn

Despite being a premium service, fiber users churn at **more than double the rate**, suggesting dissatisfaction related to pricing, expectations, or service quality.

Payment method analysis reveals behavioral insights:

- **Electronic check users:** ~45% churn
- Other methods (credit card, bank transfer): ~15–20% churn

This suggests that customers using electronic checks are **over twice as likely to churn**, possibly due to inconvenience or billing friction.

Key Business Insights (Quantified)

- ~60% churn occurs in early tenure (first few months)
- Senior citizens churn nearly **2x more** than younger customers
- Month-to-month customers churn **4–5x more** than long-term contract users
- Add-on services reduce churn risk by **~50%**
- Fiber optic users are a **high-risk segment (40%+ churn)**
- Electronic check users show the **highest churn (~45%)**

Conclusion

Customer churn is strongly influenced by **contract flexibility, service adoption, tenure, and payment behavior**. The data suggests that businesses can significantly reduce churn by:

- Strengthening **early customer engagement (first 2 months)**
- Promoting **long-term contracts with incentives**
- Encouraging **bundled services adoption**
- Improving experience for **fiber users and electronic payment customers**

Overall, implementing targeted, data-driven strategies based on these insights can potentially reduce churn by **15–25%**, leading to improved customer lifetime value and revenue stability.